

Sharan Giri

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PROFESSIONAL STATEMENT

2 years shipping production data pipelines at Barclays for Markets and Private Banking. MS in Data Science at Northeastern, building end-to-end ML systems and multi-turn agentic AI pipelines grounded in verified retrieval, with measurable reliability across latency and cost, a SIGDIAL 2026 paper under review, and production deployments on GCP and Kubernetes.

EDUCATION

Khoury College of Computer Science, Northeastern University M.S. in Data Science (GPA: 4.0) Secured Third Place at Google-Northeastern Expo for AudioSeek (Dec 2025)	Boston, MA Sep 2024 – May 2026
Sri Sivasubramaniya Nadar College of Engineering, Anna University B.E. in Computer Science Engineering (GPA: 3.7)	India Aug 2018 – Jun 2022

EXPERIENCE

Barclays Global Service Centre – Markets And Private Banking Technology Software & Automation Engineer Data Pipeline and API Integration - Built audit-ready ETL pipelines processing ML extraction outputs from unstructured email and API/JSON sources into structured formats, improving throughput by 50% and saving 85 hours/month - Enforced error volume thresholds and automated stakeholder alerting in BluePrism across 6 regulated workflows, reducing false-positive alerts by 72% - Migrated 4 UI-based automations to API-driven components, reducing turnaround time by 33% and eliminating manual dependencies across Markets and Private Banking - Owned SLA-bound production support with a 2-day resolution target, debugging live failures, coordinating controlled rollbacks and redeploying after regression testing	Aug 2022 – Jun 2024 India
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PROJECTS

CourseLENS Governed Multi-Turn Agentic Pipeline <i>SIGDIAL 2026 (Under Review)</i> - Designed a parent-child chunking pipeline mapping sparse PPTX and PDF assets into a hierarchical knowledge store - Applied curriculum-gated metadata filtering to enforce week-level content boundaries across the vector store - Architected 4-stage policy engine with deterministic routing and immutable logs, achieving 100% traceable LLM outputs - Built 90/10 hybrid retrieval as an anti-hallucination boundary, confining LLM reasoning to top-5 verified chunks - Validated 100% policy compliance via adversarial stress testing and achieved 96.15% Ragas-scored data faithfulness	Jan 2026 – Apr 2026
FinAgent Agentic Data Platform and SEC Intelligence Pipeline - Architected a two-stage agentic pipeline blending NLP extraction with generative reasoning across 72 SEC 10-K filings - Built a multi-threaded ingestion engine parallelizing numerical and textual filing data, cutting LLM token costs by 98% and latency to sub-90s - Developed a normalization layer to deduplicate XBRL tags and align 5-year time-series across 16 tracked companies - Enforced Pydantic schema validation across all LLM outputs, achieving 95%+ faithfulness on 1,500+ structured signals - Deployed FAISS retrieval over BGE-M3 embeddings with FinBERT risk scoring and FastAPI-exposed signal endpoints	Mar 2026 – Apr 2026
Early Product Success Prediction Retail Data Pipeline and Feature Engineering - Built a PySpark ETL pipeline transforming Amazon review data into a model-ready dataset for 28-day launch prediction - Engineered 40 structured signals across rating trends, sentiment polarity, and review velocity under A/B ablation - Trained multimodal XGBoost models achieving Lift@100 = 2.6x over baseline and Precision@100 = 0.65, outperforming all single-modality models - Delivered SHAP-backed explainability over final predictions enabling interpretable and auditable signal attribution	Nov 2025 – Dec 2025
AudioSEEK Audio NLP Search Platform <i>Google-Northeastern Expo</i> - Architected RAG platform converting 30+ hours of audio into a searchable knowledge base with answers under 15seconds - Built monitored ingestion with async transcription, admin-triggered restart, and email alerting on completion - Engineered hybrid SQL-FAISS store with temporal constraints to prevent lookahead bias and enforce data integrity - Deployed Dockerized GKE microservices via Terraform with MLflow for latency tracking and production drift detection	Sep 2025 – Dec 2025

TECHNICAL SKILLS

Software & Data Engineering: Python, SQL, REST APIs, FastAPI, Flask, PySpark, Pandas, Airflow, ETL/ELT Pipelines, Git, CI/CD (GitHub Actions)

AI Systems & Retrieval: RAG Pipelines, Agentic AI Systems, LangChain, LangGraph, LlamaIndex, FAISS, ChromaDB, BM25, BGE-M3, HuggingFace, FinBERT, Text Classification, Structured Output Generation

ML & Modeling: PyTorch, Scikit-learn, XGBoost, Feature Engineering, SHAP, ML Observability

MLOps & Infrastructure: Docker, Kubernetes (GKE), MLflow, DVC, Terraform

Cloud Platforms: GCP (Cloud Run, GCS, Firestore, Cloud SQL), Gemini Flash